ZINDI CHALLENGE:
FRAUD DETECTION IN ELECTRICITY AND GAS CONSUMPTION
TUNESIAN COMPANY - S.T.E.G.

Who should STEG inspect first?

The Gradient Descenders

Karl · Mohamed · Hadi · Luca

5.6 %

of STEG's customers cheat their meter.

≈ 7,500 of 135,000 – about 1 in 18 customer

10 MINUTES · 5 PARTS

01 The problem

02 The data from STEG

03 The ”leak” that fooled us

04 Baseline & our better models

05 What it means for STEG

Too many customers, too few inspectors.

1

Meters get manipulated.

Some customers make their meter show less than they use – **5.6 %** of STEG's customers. STEG loses revenue.

2

Inspections are expensive.

Every visit on site costs time and money. STEG **cannot check every customer**.

3

So: who first?

Every customer gets a fraud risk – STEG gets a **priority list** and visits the riskiest first.

135,493 customers · 4.5 M invoices (≈ 33 each)

22 → **−7** → **15**

raw columns

6 about the customer, 16 per invoice

2 ID numbers ·
1 is the answer itself ·
1 unreliable date ·
3 with no signal

features per customer

built from the invoices – one row per customer

Only **5.6 %** committed fraud – and **20 % of customers stay locked away** as our final exam.

ACCURACY LOOKS GREAT, BUT...

~~94.4 %~~

by always saying “no fraud”. That means:

it finds **zero** fraudsters.

WHAT REALLY COUNTS

Fraudsters found per 10,000 inspections (example)

Same inspection budget + more cases found = better model. 10,000 is just an example budget.

WHY ROC-AUC?

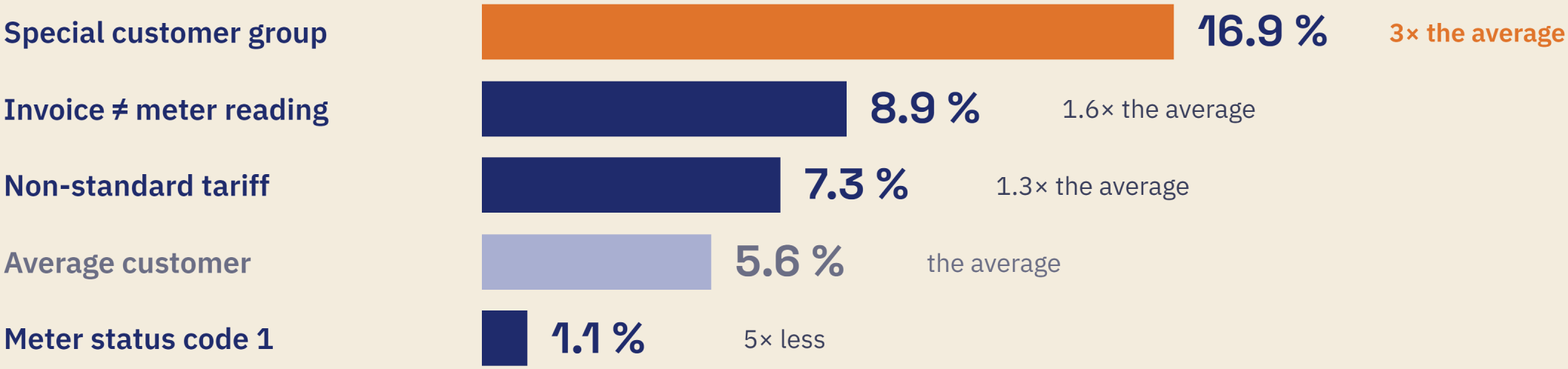
It scores the **ranking**: how often a real fraudster gets a higher risk than an honest customer. Not fooled by 94 % honest customers – and it is the Zindi metric.



0.5 coin flip **0.82 our best** 1.0 perfect

Some groups cheat 3× more often.

Signals we spotted while exploring the data. Each % = **how many customers in that group are fraudsters** – on average it is 5.6 %.



One signal alone is weak – **combined**, they point to the customers worth a visit.

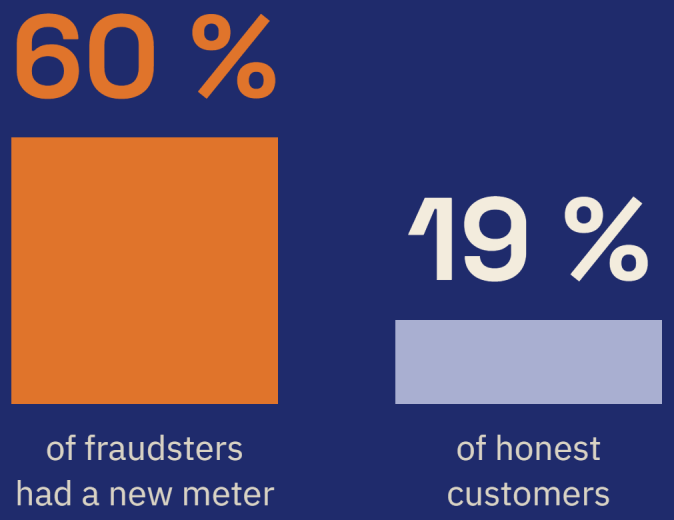
~~DATA~~ TARGET leakage

the feature: **n_counters**

(number of meters per customer)

When STEG **catches** a fraudster, it **replaces the meter**.

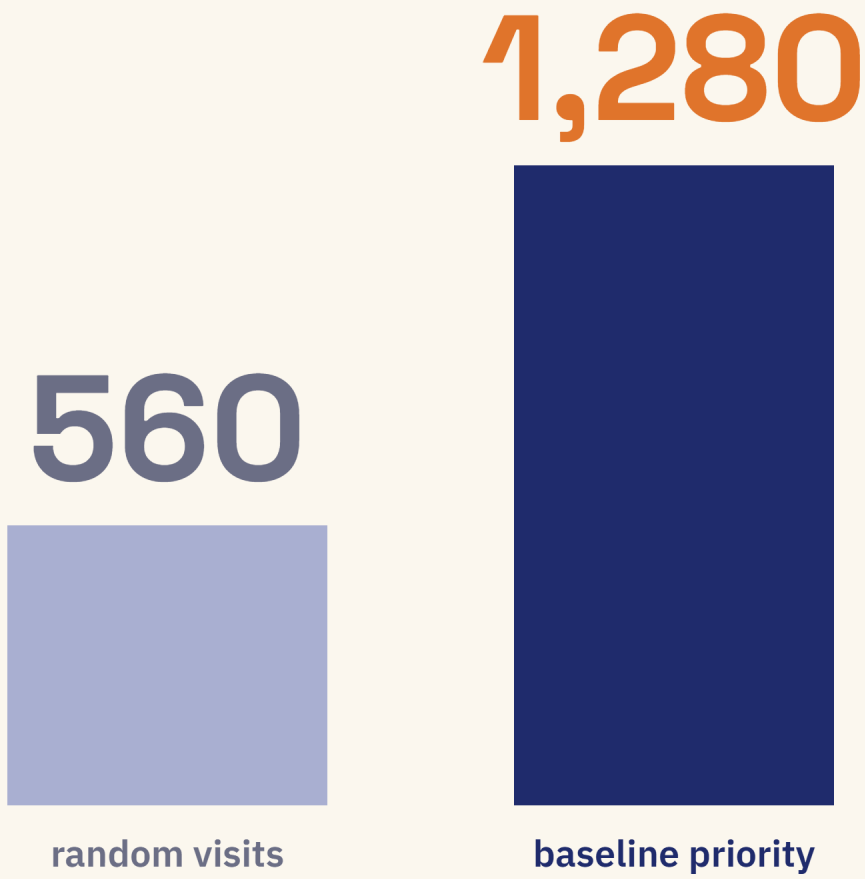
“Number of meters” was our best signal – because it was **the answer sheet**.



We removed it. Scores dropped – and became honest.

Just consumption. Already 2.3× better.

A simple **Logistic Regression** on how much each customer uses – the score every model has to beat.



Fraudsters found per 10,000 inspections (example budget) · 5-fold CV on the training part · ROC-AUC 0.663 · 8

2nd

LUCA

XGBoost

Boosted trees tuned to catch as many fraudsters as possible (F2) – now also with meter codes, like Mohamed.

AUC (CROSS-VALIDATION)

0.791

Fraudsters / 10,000* **≈ 2,210**

* 10,000 = example budget · AUC = 5-fold CV · ** Hadi: estimate · 9

1st

MOHAMED

LightGBM

Boosted trees with location and 30+ new invoice features, tuned.

AUC (CROSS-VALIDATION)

0.819 after tuning

↑ **0.747** at the start

Fraudsters / 10,000* **≈ 2,450**

3rd

HADI

Ensemble

Five models voting together: boosting, random forest, tree, KNN, logistic regression.

AUC (CROSS-VALIDATION)

0.70

Fraudsters / 10,000* **≈ 1,350****

From 0.747 to 0.819, step by step.

1

16 base features

Consumption, zero bills, invoice ≠ meter, tariff, customer group.

CV AUC
0.747

2

+ Location

District and region of the customer.

CV AUC
0.768

3

+ Usage pattern

How steady it is, drops over time, summer vs. winter.

CV AUC
0.774

4

+ Bills & meter

Gaps between bills, electricity vs. gas, meter codes, customer age.

CV AUC
0.817

5

+ Tuning

Settings tested on the training part only. Holdout: 0.821.

CV AUC
0.819

CV AUC = 5-fold CV on the training part · holdout = 27,099 unseen customers · 10

05 OF 05 – WHAT IT MEANS FOR STEG

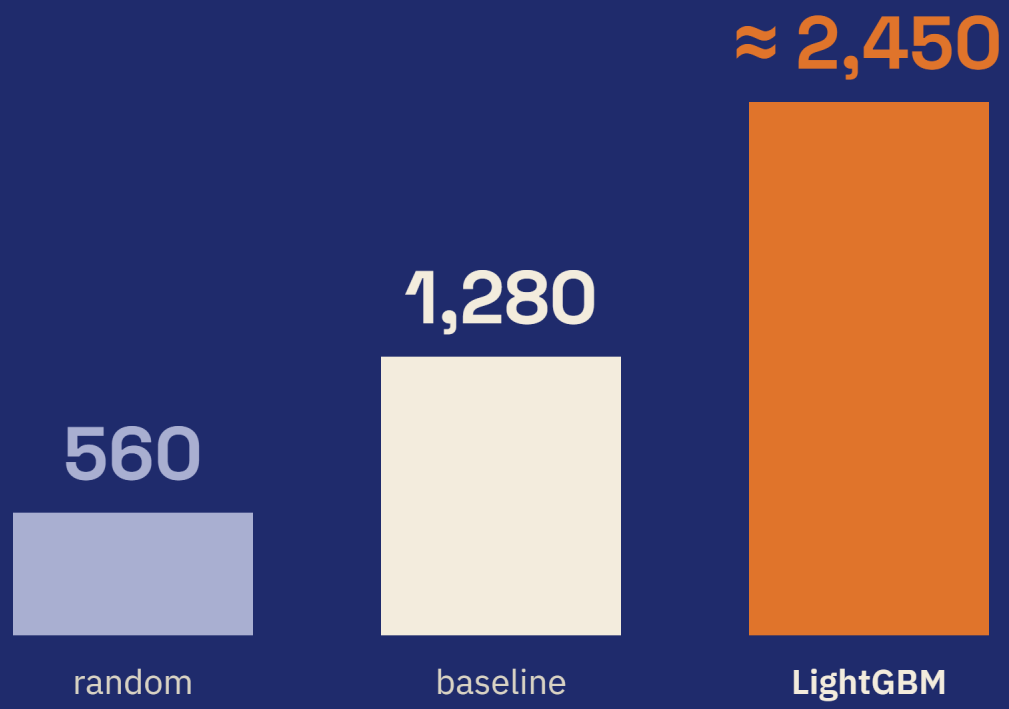
4.4×

the fraudsters vs. random visits. Same inspections.

5.6 %
hit rate, random visit

↑

24.5 %
hit rate with our model (riskiest 10 %)



Per 10,000 inspections (example budget) · 5-fold CV · 11

Our product: a **monthly priority list** for the field teams.
Here is what held it back – and what would help.

WHAT WE DIDN'T HAVE

- What do some columns mean?
- When was the fraud detected?
- Is using location fair?

WHAT WOULD MAKE IT BETTER

- STEG stores the detection date
- Smart meters: real-time data
- Seasonal and trend signals

Thank you.

Same budget. More fraud found. Honestly measured.

Questions?

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